YouTube Big Data Analytics Pipeline - Deployment Manual

Table of Contents

1. System Requirements
2. Environment Setup
3. Data Collection
4. Data Cleaning
5. Storage Configuration
6. Processing Pipeline
7. Analytics Execution
8. Visualization
9. Troubleshooting

1. System Requirements

Minimum Requirements

- **GCP Account** with billing enabled
- **Local System**: 8GB RAM, 50GB free disk space
- **Python**: 3.8 or higher
- **Browser**: Chrome/Firefox (latest version)
- **Internet**: Stable connection (minimum 10 Mbps)

Required GCP Services

- Cloud Storage
- Dataproc
- BigQuery
- Cloud Shell

2. Environment Setup

2.1 Initial GCP Configuration

```
# Set up GCP project
gcloud config set project YOUR_PROJECT_ID

# Enable required APIs
gcloud services enable dataproc.googleapis.com
gcloud services enable storage.googleapis.com
gcloud services enable bigquery.googleapis.com

# Set default region
gcloud config set dataproc/region us-central1

# Create storage buckets
gsutil mb -p YOUR_PROJECT_ID -l us-central1 gs://YOUR_PROJECT_ID-storage/
gsutil mb -p YOUR_PROJECT_ID -l us-central1 gs://YOUR_PROJECT_ID-staging/
```

2.2 Local Environment Setup

```
# Install required packages
pip install --upgrade pip
pip install pandas numpy kaggle google-cloud-storage google-cloud-bigquery
pip install pyspark pyarrow plotly matplotlib seaborn
```

2.3 Authentication Setup

Authenticate GCP

gcloud auth login

gcloud auth application-default login

Set up Kaggle API

mkdir -p ~/.kaggle

echo '{"username":"YOUR_KAGGLE_USERNAME","key":"YOUR_KAGGLE_KEY"}' > ~/.kaggle/kaggle.json

chmod 600 ~/.kaggle/kaggle.json

3. Data Collection

3.1 Directory Structure Creation

```
# Create project structure
mkdir -p ~/youtube_pipeline/{data/{raw,processed},scripts,config,outputs}
cd ~/youtube_pipeline
```

3.2 Kaggle Data Download

```
# Download YouTube trending data
cd ~/youtube_pipeline/data/raw
kaggle datasets download datasnaek/youtube-new
unzip youtube-new.zip
rm youtube-new.zip

# Verify downloaded files
ls -la *.csv
# Expected: USvideos.csv, GBvideos.csv, etc. (10 country files)
```

3.3 YouTube API Data Collection (Optional)

```
# Create API configuration
cat > ~/youtube_pipeline/config/api_config.py << 'EOF'
YOUTUBE_API_KEYS = ['YOUR_YOUTUBE_API_KEY']
EOF

# Run collection script
python3 scripts/youtube_api_collector.py
```

3.4 Upload Raw Data to GCS

Upload to Cloud Storage

gsutil -m cp -r ~/youtube_pipeline/data/raw/*.csv gs://YOUR_PROJECT_ID-storage/raw/

4. Data Cleaning

4.1 Execute Cleaning Pipeline

```
cd ~/youtube_pipeline

# Run cleaning script
python3 scripts/data_cleaner.py

# Verify cleaned data
ls -la data/processed/
# Expected: kaggle_youtube_cleaned.csv (~290MB)
```

4.2 Upload Cleaned Data

Upload processed data to GCS

gsutil cp data/processed/kaggle_youtube_cleaned.csv gs://YOUR_PROJECT_ID-storage/processed/

5. Storage Configuration

5.1 Create Dataproc Cluster

```
# Create cluster for big data processing
gcloud dataproc clusters create youtube-analytics-cluster \
    --region=us-central1 \
    --zone=us-central1-a \
    --master-machine-type=n1-standard-4 \
    --worker-machine-type=n1-standard-4 \
    --num-workers=2 \
    --image-version=2.0 \
    --optional-components=JUPYTER \
    --enable-component-gateway \
    --project=YOUR_PROJECT_ID
```

5.2 Create BigQuery Dataset

```
# Create BigQuery dataset
bq mk --location=us-central1 --dataset YOUR_PROJECT_ID:youtube_analytics
```

5.3 Execute Storage Pipeline

```
# Upload storage script
gsutil cp scripts/storage_pipeline.py gs://YOUR_PROJECT_ID-storage/scripts/

# Submit storage job
gcloud dataproc jobs submit pyspark \
    --cluster=youtube-analytics-cluster \
    --region=us-central1 \
    gs://YOUR_PROJECT_ID-storage/scripts/storage_pipeline.py

# Monitor job
gcloud dataproc jobs list --cluster=youtube-analytics-cluster --region=us-central1
```

6. Processing Pipeline

6.1 Batch Processing

```
# Upload processing script
gsutil cp scripts/batch_processing.py gs://YOUR_PROJECT_ID-storage/scripts/

# Execute batch processing
gcloud dataproc jobs submit pyspark \
    --cluster=youtube-analytics-cluster \
    --region=us-central1 \
    --properties="spark.sql.adaptive.enabled=true" \
    gs://YOUR_PROJECT_ID-storage/scripts/batch_processing.py
```

6.2 Stream Processing (Optional)

```
# For real-time processing
gcloud dataproc jobs submit pyspark \
    --cluster=youtube-analytics-cluster \
    --region=us-central1 \
    gs://YOUR_PROJECT_ID-storage/scripts/stream_processing.py
```

7. Analytics Execution

7.1 Topic Analysis

```
# Upload analytics script
gsutil cp scripts/topic_analysis.py gs://YOUR_PROJECT_ID-storage/scripts/

# Run analytics for 4 research topics
gcloud dataproc jobs submit pyspark \
    --cluster=youtube-analytics-cluster \
    --region=us-central1 \
    gs://YOUR_PROJECT_ID-storage/scripts/topic_analysis.py
```

7.2 Verify Results in BigQuery

```
# Query results
bq query --use_legacy_sql=false \
'SELECT * FROM `YOUR_PROJECT_ID.youtube_analytics.viral_analysis` LIMIT 10'

# Export results to GCS
bq extract --destination_format=CSV \
    YOUR_PROJECT_ID:youtube_analytics.viral_analysis \
    gs://YOUR_PROJECT_ID-storage/outputs/viral_analysis.csv
```

8. Visualization

8.1 Local Setup (Windows/Mac/Linux)

```
# Install Jupyter and dependencies
pip install jupyter notebook
pip install plotly pandas numpy matplotlib seaborn kaleido

# Start Jupyter Notebook
jupyter notebook
```

8.2 Download Analysis Results

```
# Create local directory
mkdir ~/youtube_visualizations

# Download results from GCS
gsutil -m cp -r gs://YOUR_PROJECT_ID-storage/outputs/*.csv ~/youtube_visualizations/
```

8.3 Run Visualization Notebook

1. Open Jupyter Notebook
2. Create new Python 3 notebook
3. Copy and run visualization code from visualization.ipynb
4. Execute all cells sequentially

8.4 Export Visualizations

The notebook will automatically save:

- 15 PNG files (high-resolution images)
- 15 HTML files (interactive charts)
- 1 PDF report (optional)

Output location: ~/youtube_visualizations/

Troubleshooting

Common Issues and Solutions

Issue 1: Kaggle API Authentication Failed

Solution: Regenerate Kaggle API token

Go to https://www.kaggle.com/account → Create New API Token

Issue 2: Dataproc Job Failed

```
# Check logs
gcloud dataproc jobs describe JOB_ID --region=us-central1

# View detailed logs
gsutil cat gs://YOUR_PROJECT_ID-staging/.../driveroutput
```

Issue 3: Out of Memory Error

```
# Increase cluster resources
gcloud dataproc clusters update youtube-analytics-cluster \
    --num-workers=3 \
    --region=us-central1
```

Issue 4: Visualization Not Rendering

```
# Install required renderer
pip install -U kaleido
# Or use online mode
import plotly.io as pio
pio.renderers.default = 'browser'
```